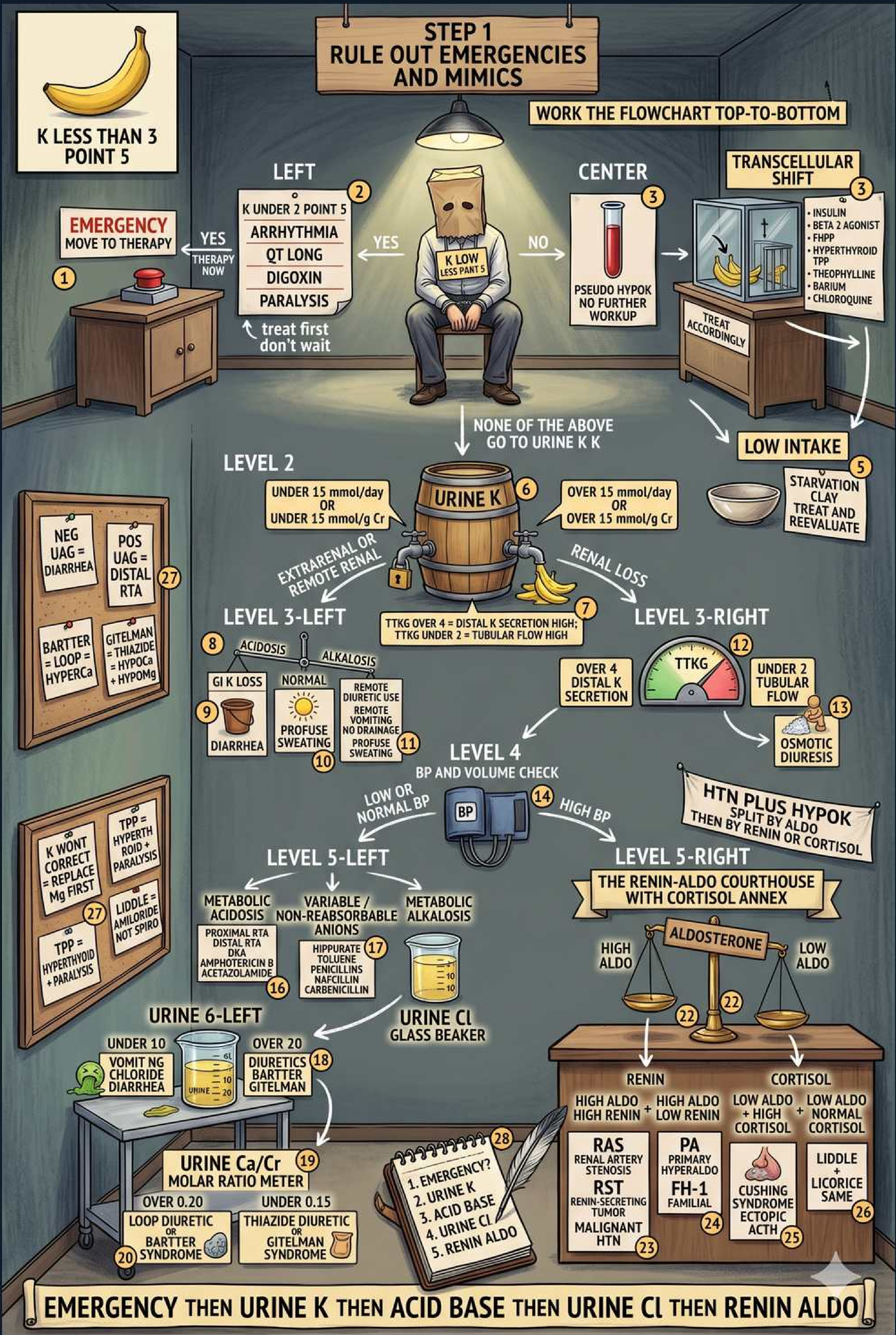
Hypokalemia — diagnosis algorithm



1. Step 1: rule out emergencies/mimics

WHAT YOU SEE

Top banner: STEP 1 rule out emergencies and mimics; work flowchart top-to-bottom

WHAT IT ENCODES

Fixed algorithm order: emergency → transcellular shift / low intake → urine K → acid-base (and BP) → urine Cl → renin/aldo. Always work top-down so you don't order an expensive renin/aldo before checking the cheap discriminators. Hypokalemia = serum K <3.5 mmol/L; <2.5 is severe and earns ECG monitoring.

BOARD TRAP

Board distractor: ordering renin/aldosterone first in a hypokalemic patient. The correct first moves are an ECG and checking for shift/intake — renin/aldo only matter much later, after urine K confirms renal loss AND the patient is hypertensive.

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2. Emergency → treat first

WHAT YOU SEE

Left: EMERGENCY move to therapy; arrhythmia, QT long, paralysis; treat first don't wait

WHAT IT ENCODES

Life-threatening features — ventricular arrhythmia, long QT/U waves, flat T waves, ascending paralysis, digoxin use — mandate immediate IV K replacement before completing the workup. IV KCl no faster than 10–20 mmol/hr peripherally (up to 40 mmol/hr central with monitoring); each 10 mmol raises serum K ~0.1 mmol/L. Always correct Mg simultaneously.

BOARD TRAP

Giving K in dextrose-containing fluid (D5W) in an emergency: the dextrose triggers insulin release that shifts K INTO cells and transiently WORSENS the hypokalemia. Use saline as the diluent for urgent K repletion.

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3. Transcellular shift

WHAT YOU SEE

Right: TRANSCELLULAR SHIFT — insulin, beta2, alkalosis, DKA, refeeding, catecholamines

WHAT IT ENCODES

Center branch: exclude redistribution before chasing a leak. Insulin, beta-2 agonists (albuterol), alkalosis, catecholamines (acute MI, head injury), refeeding, and periodic paralysis all drive K into cells via the Na/K-ATPase, so total body K is normal. Serum K low without a true deficit.

BOARD TRAP

Treating a pure shift (e.g. thyrotoxic or familial periodic paralysis) with aggressive KCl. Once the trigger resolves K floods back out → REBOUND hyperkalemia. Pseudohypokalemia (very high WBC absorbing K in the tube) similarly has no real deficit — repeat on a quickly-processed sample.

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4. Low intake

WHAT YOU SEE

LOW INTAKE: starvation, treat and re-evaluate

WHAT IT ENCODES

Poor intake (starvation, alcoholism, clay/pica) alone rarely causes significant hypokalemia because the healthy kidney can lower urine K to <15 mmol/day. It is usually a contributor that unmasks a coexisting shift or loss. Replace and re-evaluate.

BOARD TRAP

Attributing marked hypokalemia to diet alone. If urine K is inappropriately high (>15–20) despite a starvation story, there is a renal leak (often a hidden diuretic or eating-disorder vomiting) — keep looking.

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5. Level 2: urine K split

WHAT YOU SEE

URINE K barrel; under 15 mmol/day or under 13 mmol/g Cr = extrarenal; over 15 = renal loss

WHAT IT ENCODES

Level 2 is the master fork: 24-hr urine K <15 mmol/day (or spot <13 mmol/g Cr, or TTKG <3) = appropriate renal conservation → extrarenal/GI loss or shift. >15 mmol/day = inappropriate renal wasting. This single test sends you down the left (GI) or right (renal) arm of the whole tree.

BOARD TRAP

Interpreting a spot urine K alone without correcting for urine concentration. A concentrated random sample can read falsely high; use mmol/g creatinine or TTKG, and never read urine K in isolation from acid-base status and BP.

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6. Extrarenal / diarrhea

WHAT YOU SEE

Level 3-left: low urine K; diarrhea, profuse sweating, remote diuretic

WHAT IT ENCODES

Low-urine-K (extrarenal) branch: diarrhea, laxative abuse, villous adenoma/VIPoma, profuse sweating, and REMOTE (days-old) diuretic use after the drug has cleared. Diarrhea brings a non-anion-gap metabolic acidosis with a NEGATIVE urine anion gap.

BOARD TRAP

A laxative abuser or bulimic faking is the classic low-urine-K mimic. They look like distal RTA (both hypokalemic), but diarrhea gives a NEGATIVE urine anion gap whereas distal RTA gives a POSITIVE one — that sign discriminates them.

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7. TTKG / distal secretion

WHAT YOU SEE

Level 3: TTKG over 4 + distal secretion; tubular flow high

WHAT IT ENCODES

Transtubular K gradient = (urine K/plasma K) ÷ (urine osm/plasma osm); it estimates aldosterone-driven distal K secretion. With hypokalemia, TTKG >4 indicates INAPPROPRIATE renal K secretion (a leak); <2–3 is appropriate conservation. Valid only when urine osm > serum and urine Na >25.

BOARD TRAP

Computing TTKG when urine is dilute or urine Na is low — the formula assumes ADH-driven water reabsorption and adequate distal Na delivery, so it is invalid otherwise and will mislead.

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8. Osmotic diuresis

WHAT YOU SEE

Level 3-right: TTKG, distal secretion, osmotic diuresis

WHAT IT ENCODES

On the renal-wasting arm, increased distal tubular FLOW is itself a driver: osmotic diuresis (hyperglycemia in DKA, mannitol, high-protein/urea load) sweeps K out via flow-dependent ROMK secretion, independent of aldosterone. BP is usually normal.

BOARD TRAP

In DKA the osmotic diuresis plus insulin deficiency masks a profound TOTAL-body K deficit even when serum K is normal or high — once insulin is started, K crashes. Add K to fluids when serum K <5.2 and hold insulin if K <3.3.

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9. Level 4: BP + volume check

WHAT YOU SEE

LEVEL 4: BP and volume check; low/normal BP vs high BP

WHAT IT ENCODES

Level 4 splits the renal-loss arm by blood pressure. Normal/low BP → tubulopathy or diuretic (Bartter, Gitelman, vomiting, loop/thiazide). HIGH BP → mineralocorticoid excess (Conn, renovascular, Cushing, Liddle, licorice). BP is the gate BEFORE renin/aldo.

BOARD TRAP

Skipping the BP check and going straight to renin/aldo. Renin/aldo are only interpretable once HTN is established; in a normotensive hypokalemic patient the answer is a tubulopathy or diuretic, and renin/aldo waste money.

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10. Level 5-left: acid-base

WHAT YOU SEE

LEVEL 5-LEFT: metabolic acidosis vs variable vs metabolic alkalosis

WHAT IT ENCODES

Normal-BP branch is sorted by acid-base. Metabolic ACIDOSIS → type 1 (distal) or type 2 (proximal) RTA, or toluene. Metabolic ALKALOSIS → vomiting, diuretics, Bartter, or Gitelman. Variable → check the next test (urine Cl, urine Ca).

BOARD TRAP

Forgetting that RTA causes hypokalemia WITH acidosis while almost every other normotensive cause (vomiting, diuretics, Bartter, Gitelman) causes hypokalemia WITH alkalosis. Pairing acidosis with Bartter/Gitelman is a classic wrong answer.

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11. Level 6: urine Ca/Cr

WHAT YOU SEE

LEVEL 6-LEFT: urine Ca/Cr molar ratio meter; over 0.20 loop/Bartter, under 0.15 Gitelman/thiazide

WHAT IT ENCODES

Within the normotensive-alkalosis tubulopathies, the urine Ca:Cr molar ratio separates the two: HIGH (>0.20) = hypercalciuria of LOOP diuretics / Bartter (NKCC2 block kills the lumen-positive voltage that drives Ca reabsorption). LOW (<0.15) = hypocalciuria of THIAZIDE / Gitelman (NCC block enhances proximal Ca reabsorption).

BOARD TRAP

Mixing up the calcium signatures. Gitelman = thiazide-like = HYPOcalciuria + HYPOmagnesemia; Bartter = loop-like = HYPERcalciuria (with nephrocalcinosis). Calling Bartter hypocalciuric is the trap.

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12. Urine Cl beaker

WHAT YOU SEE

URINE Cl glass beaker (saline-responsive vs resistant alkalosis)

WHAT IT ENCODES

In metabolic alkalosis use urine CHLORIDE (not Na) to judge volume. Urine Cl <20 = saline-RESPONSIVE: vomiting/NG suction or remote diuretic (volume deplete) — corrects with normal saline. Urine Cl >20 = saline-RESISTANT: active diuretic, Bartter/Gitelman, or mineralocorticoid excess.

BOARD TRAP

Using urine Na to assess volume in a vomiting patient. Bicarbonaturia obligates Na excretion so urine Na is falsely HIGH despite true volume depletion — urine Cl stays low and is the reliable marker.

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13. Level 5-right: renin-aldo court

WHAT YOU SEE

LEVEL 5-RIGHT: renin-aldo courthouse with cortisol annex; high aldo vs low aldo

WHAT IT ENCODES

High-BP branch: the renin/aldosterone (± cortisol) pattern names the disease. Aldo high + renin low = primary hyperaldosteronism (Conn). Aldo high + renin high = secondary (renovascular, malignant HTN). Aldo low + renin low = non-aldo mineralocorticoid effect (Liddle, licorice, Cushing).

BOARD TRAP

Reflexively diagnosing Conn whenever HTN + hypokalemia appear. If BOTH renin and aldo are low, the cause is cortisol/ENaC-driven (Cushing, licorice, Liddle), not an aldosterone-secreting adenoma — the aldo:renin ratio must be interpreted, not assumed.

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14. HTN renin/aldo patterns

WHAT YOU SEE

RAS/RST high renin; PA/Cushing patterns; Liddle, Liddle same low-low

WHAT IT ENCODES

Pattern cheat-sheet for HTN + hypoK: renovascular/renin-secreting tumor = renin↑ aldo↑; primary aldosteronism (adenoma/bilateral hyperplasia) = renin↓ aldo↑ (aldo:renin ratio >20–30); Liddle, licorice/AME, and Cushing = renin↓ aldo↓ (then split by cortisol). Confirm primary aldo with a saline-suppression test.

BOARD TRAP

Picking spironolactone for Liddle syndrome. Liddle is a gain-of-function ENaC mutation (low aldo), so an aldosterone-receptor blocker fails — the answer is amiloride or triamterene, which block the channel directly.
